



United International University (UIU)
Dept. of Computer Science and Engineering (CSE)
Mid-Term Exam Year: **2025** Semester: **Summer**
Course Code: **CSE 3811** Title: **Artificial Intelligence**
Marks: **30** Time: **1 Hour 30 minutes**

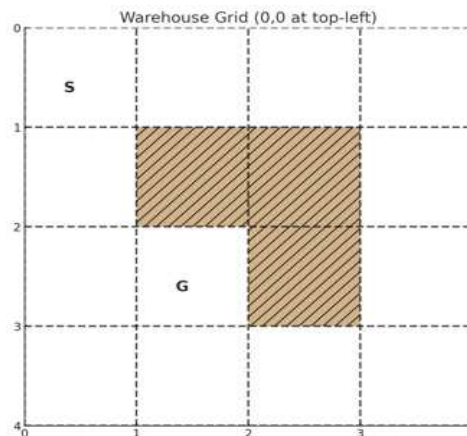
Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Answer all the questions. All questions are of values indicated on the right-hand margin.

1. A space agency has developed an autonomous lunar exploration rover named "LunaX". The rover's mission is to explore the Moon's surface, collect rock samples, and transmit scientific data to Earth. It must navigate challenging terrain, avoid hazards such as craters, and operate efficiently with limited energy resources.

- a. Write the **PEAS description** for the "LunaX" rover. [2]
- b. Suppose the rover detects several rock formations on the lunar surface. The rover must plan a path to collect all of them efficiently. Formulate this as a search problem using the **Problem Formulation** steps. [3]

2. You are designing a route-planning software that helps a delivery robot find the shortest path inside a warehouse. The warehouse layout is represented as a grid where **each cell is either free or blocked by an obstacle**. The robot can move **up, down, left, or right** (only if the adjacent cell is free), and each move has a cost. You formulate the search problem in this manner:



State: Current location of the robot in the grid.

Start state: The robot's starting cell.

Successor Function: Moves to any free adjacent cell (up, down, left, right).

Path Cost: Each move has a cost of 1 unit.

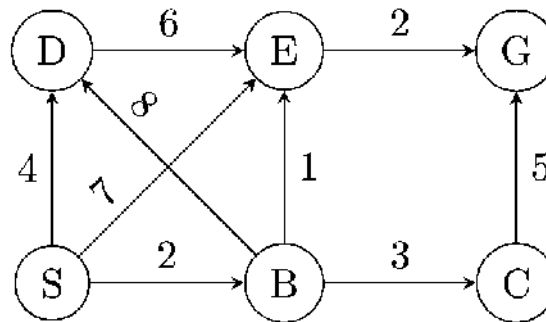
Goal Test: The robot reaches the target cell where the package is located.

Now, consider the following scenario: The robot starts at cell **S (0,0)** and needs to reach **Goal G (2,1)**. The **blocked cells are: (1,1), (1,2), (2,2)**.

Assume that **tie-breaking is done in the following manner: Up > Down > Right > Left, whenever multiple options exist**. Also assume the heuristic used for A* is the Manhattan distance. Your search algorithm will try to expand nodes until the goal is found.

- a. If a node is not expanded twice, State which path will be returned by:
 - i. **BFS**
 - ii. **A* Graph Search**
 - iii. **Depth Limited Search (DLS) with limit = 5** [3x1=3]
- b. If the cost of moving along the x-axis is defined to be four times the cost of moving along the y-axis, which path would the Uniform Cost Search (UCS) algorithm return? [2]

3. Consider the following graph where **S** is the start node and **G** is the goal node.



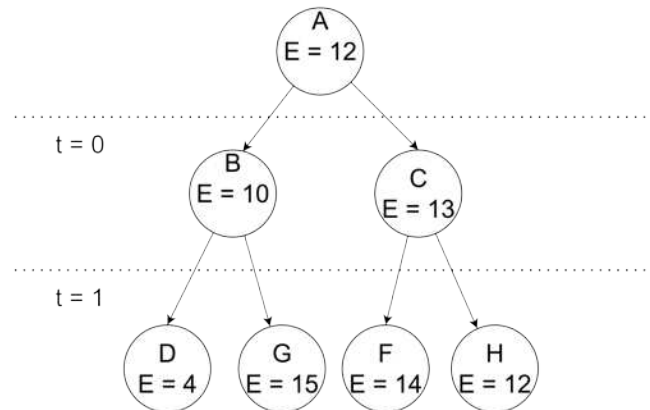
- a. Assign **admissible heuristic values** for each node of the graph with necessary calculations showing that the heuristics are, in fact, admissible. [2]
- b. You are now given a **new set of heuristic values** which are **all dominant** over the ones you have assigned in (a) and admissible as well. What is the **merit and the demerit** of this new set of heuristic values compared to your ones? [1]
- c. For any random graph, you are given 4 different heuristic functions: $h_1(n)$, $h_2(n)$, $h_3(n)$ and $h_4(n)$. The values that $h_1(n)$, $h_2(n)$ and $h_3(n)$ give for any state of the graph are always **admissible** but the values from $h_4(n)$ are always **inadmissible**. Now we create two new heuristic functions: $h_5(n)$ and $h_6(n)$ using the previous functions as below:

$$h_5(n) = \max(h_1(n), h_4(n)) + \max(h_2(n), h_4(n)) - 2 \times h_4(n) + h_3(n)$$

$$h_6(n) = \text{average}(h_2(n), h_3(n), h_4(n)) - \frac{h_4(n)}{3} - \frac{2 \times h_2(n)}{3}$$

Determine if $h_5(n)$ and $h_6(n)$ are **admissible or not**. Write the calculation or logic behind your answer. [2]

4. Consider the following figure for **Simulated Annealing** on a **maximization problem**:



Above is a simple search tree where A is the current node and G is the target node. All nodes have their own **utility/energy (E)**. The **temperature function** is given by:

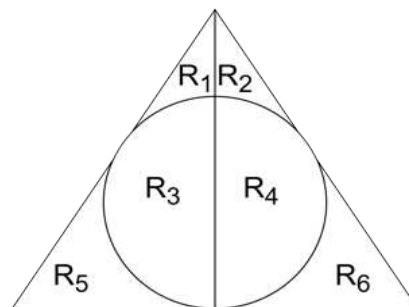
$T = 8 - 2 \cdot t$; where t = timepoint

The neighbor / child nodes are always checked from left to right. Consider that all the nodes use the value of **0.5 as the random number generator (RNG)** that may be checked against the threshold. [Hint: Probability for taking a bad move is $e^{\Delta E/T}$, here: $\Delta E = E(\text{neighbor}) - E(\text{current})$]

Answer the following questions:

- If we apply basic **Hill Climbing**, which node will we reach in the end? Will you encounter any problems? [2]
- When applying **Simulated Annealing**, which node will we reach in the end? [3]

5. Consider the following “Deathly Hallows” symbol consisting of **6 regions**. You need to fill each region with one of the **4 colors: Red, Blue, Green, or Yellow**, in such a way that all 6 regions are distinguishable after assigning the colors.



Some additional constraints:

- Region 1 cannot use Green or Yellow.
- Region 2 must be colored Red.
- Region 5 cannot use Yellow.
- Region 3 and Region 6 must not share the same color.
- Region 5 must be different from Region 6.
- Red can be used exactly 2 times.

Now formulate the problem as a **Constraint Satisfaction Problem (CSP)**. You must apply both **MRV** and **LCV** heuristics while choosing and assigning a color to a tile. Show the steps. [5]
 [Hint: Do not assign same color to adjacent tiles]

6. Consider the following game tree. The children of each node must be visited **from right to left** during the minimax search.

a. Determine the ranges of values for a, b, c, and d such that the highlighted branches are pruned when applying the minimax algorithm with alpha-beta pruning. Also, indicate the **utility value** at each internal node. [4]

b. Provide a brief explanation of why one of the pruned branches is cut off under your chosen range of values (e.g., justify the range of $a \in [x, y]$). [1]

